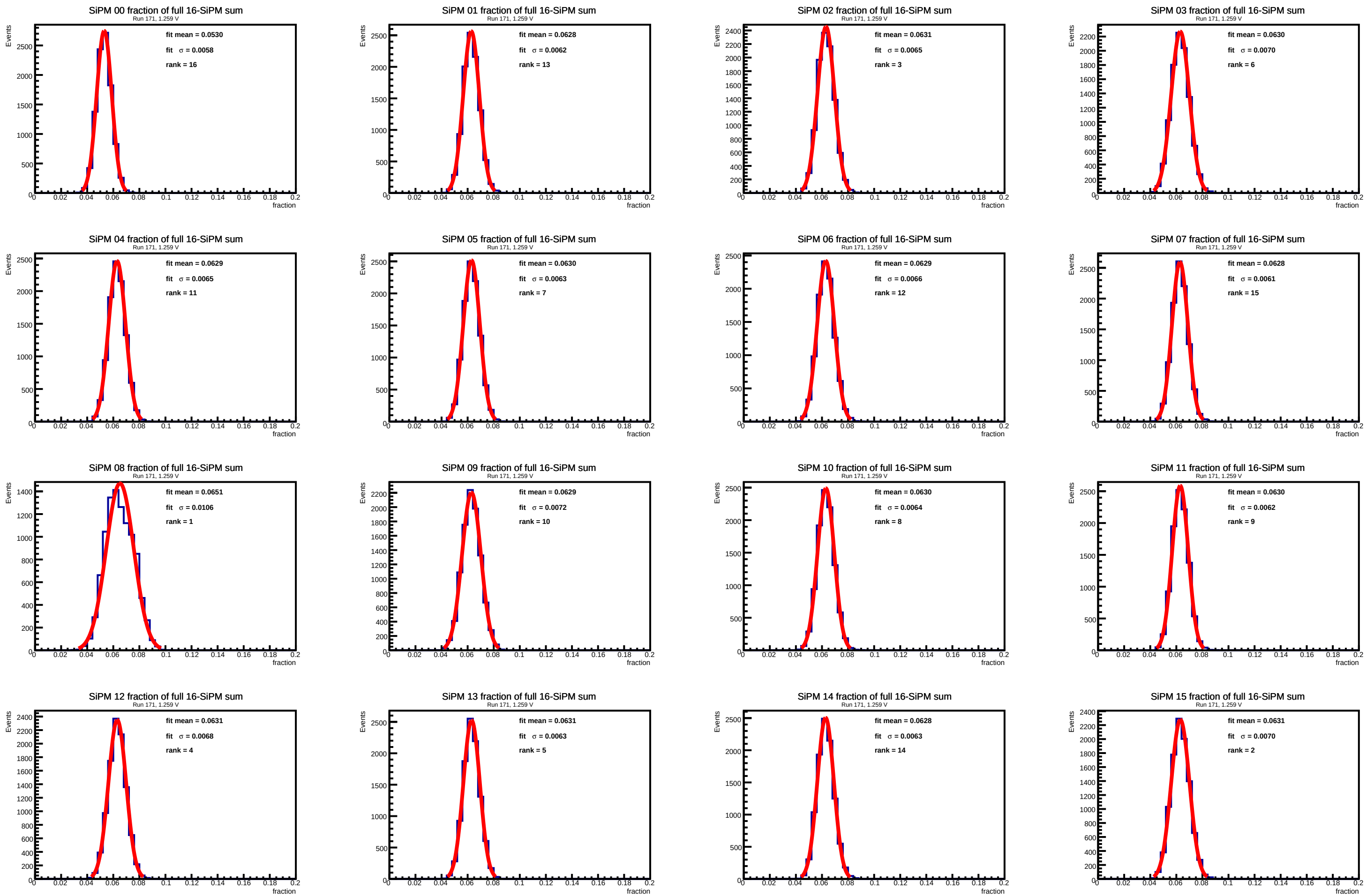
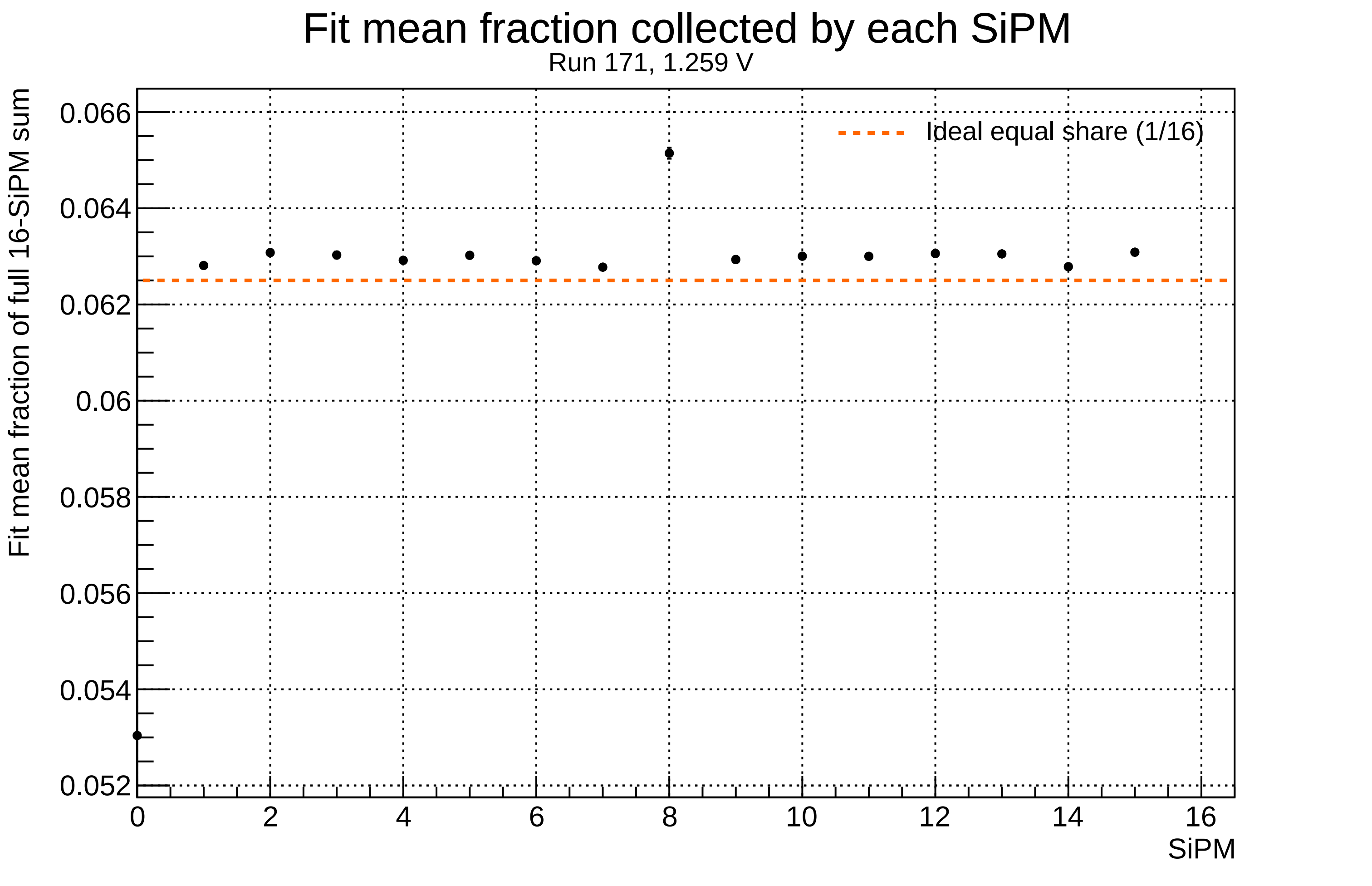
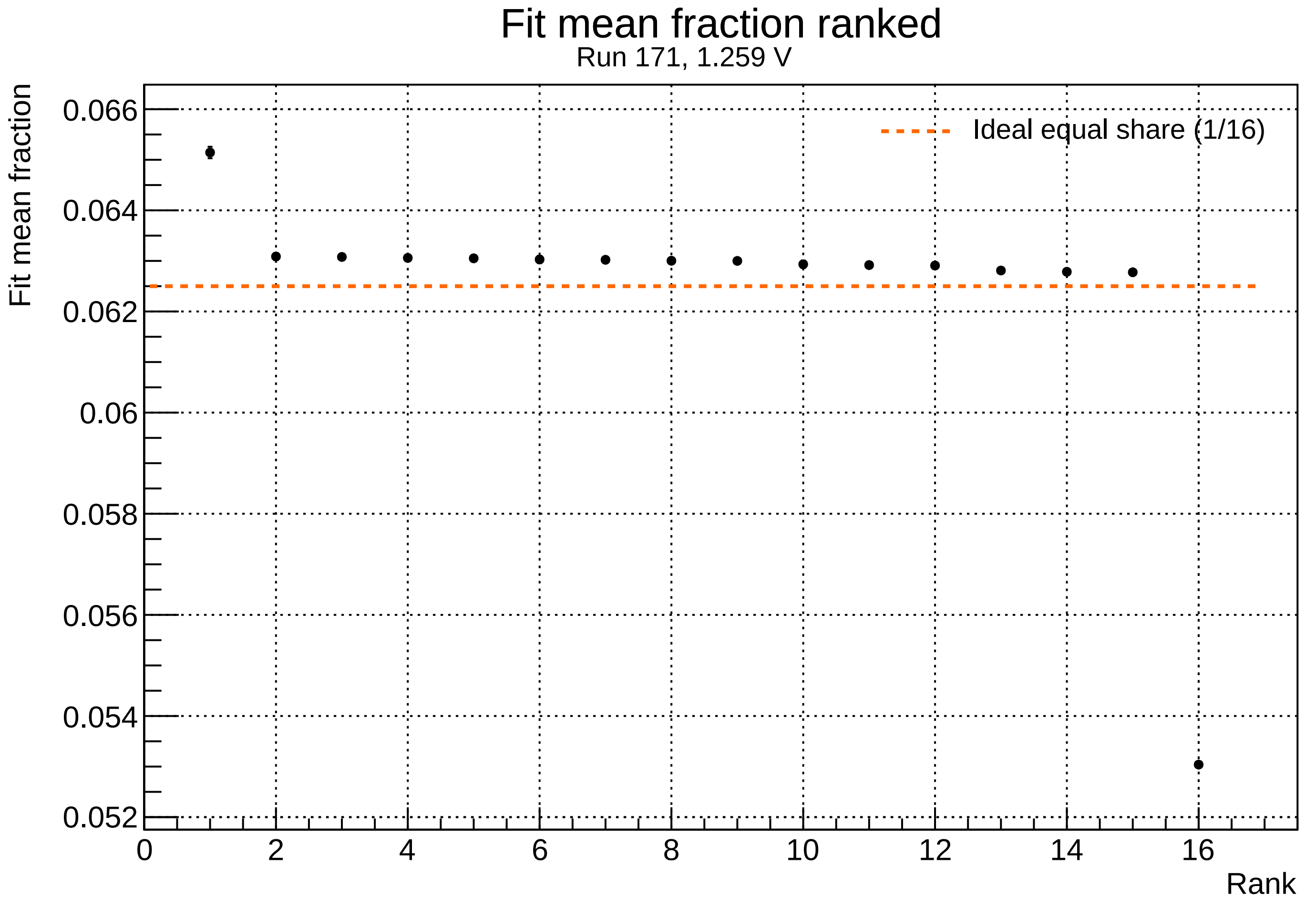
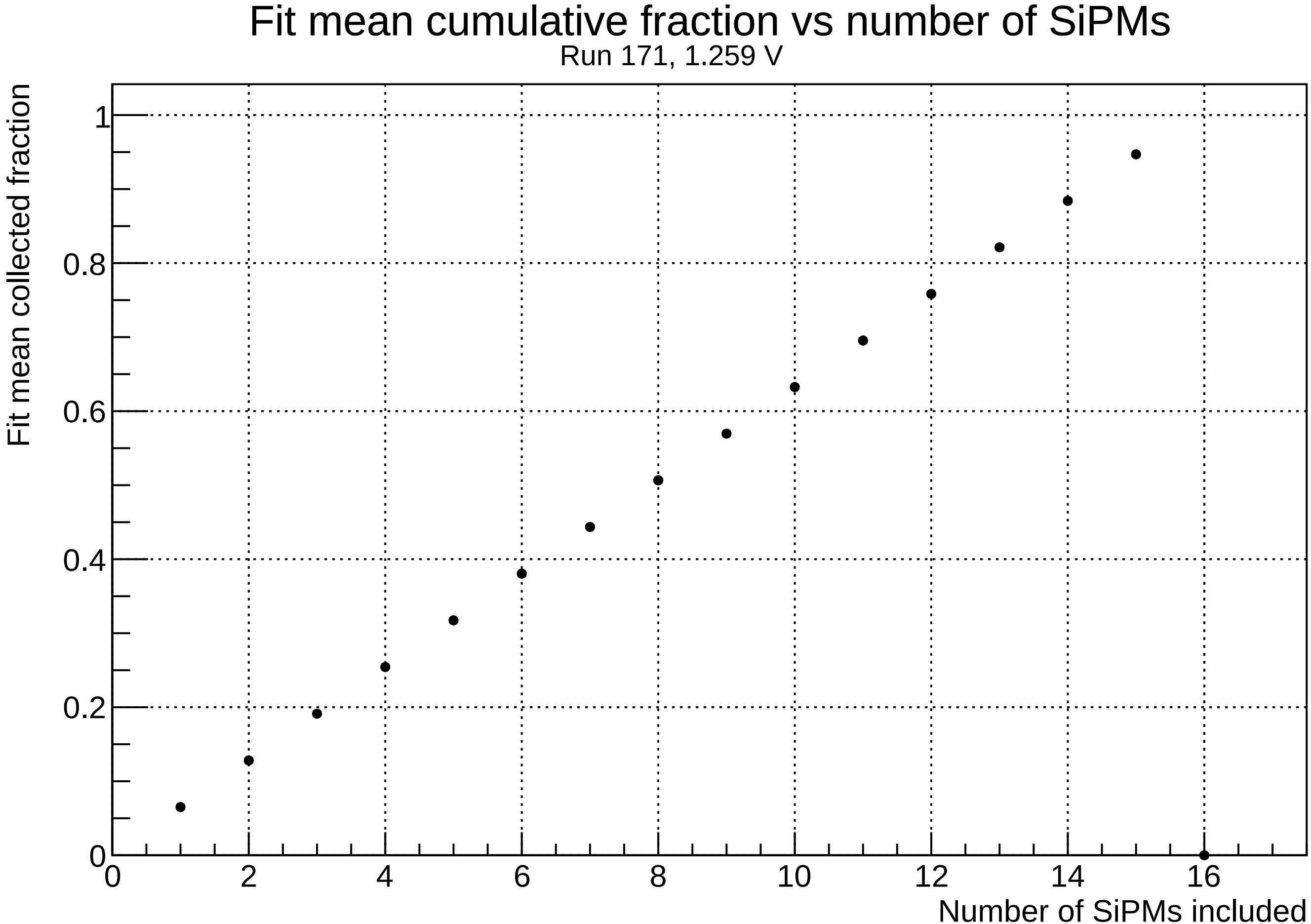
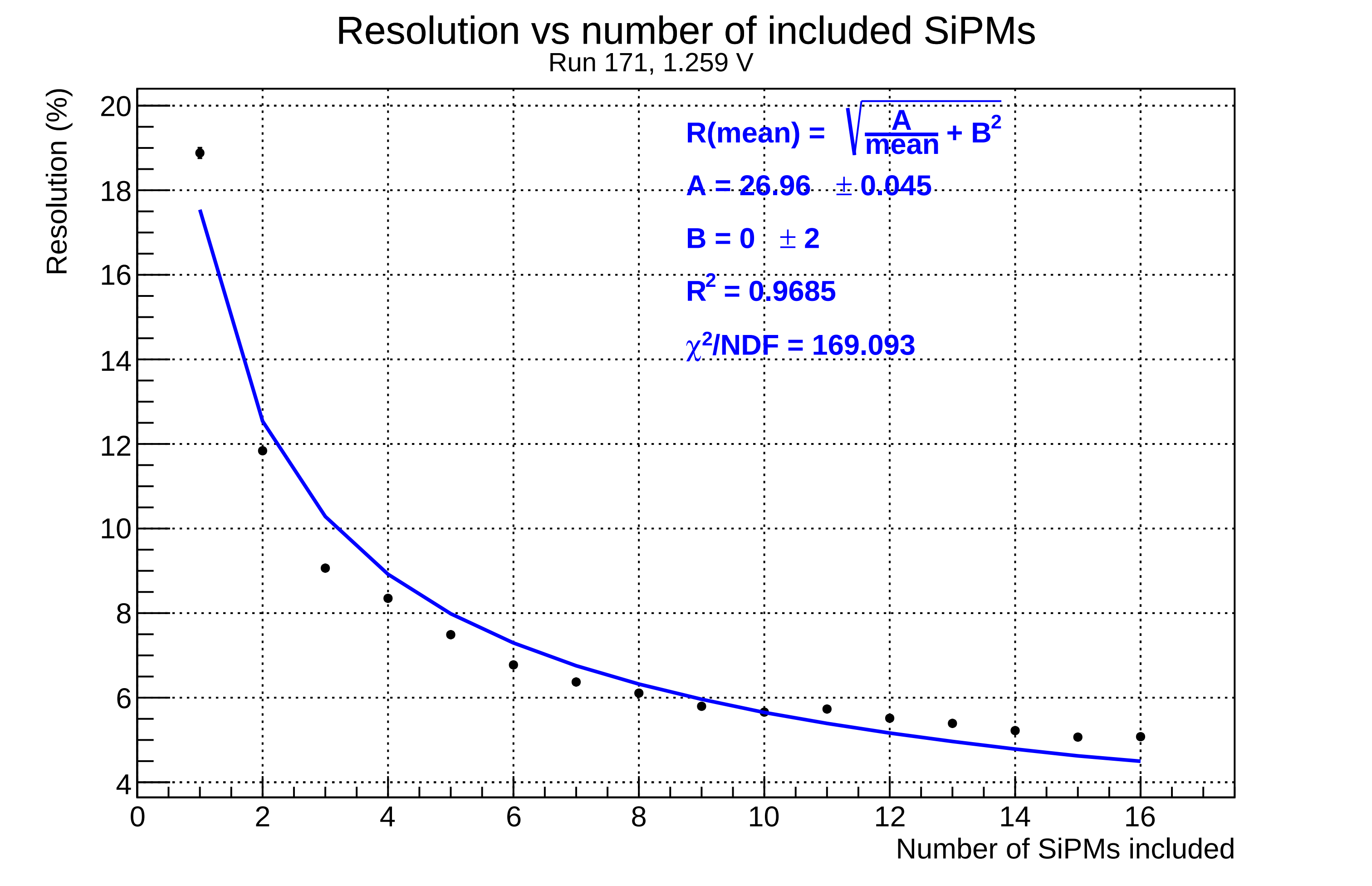

SiPM 00 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0530
fit σ = 0.0058
rank = 16
SiPM 01 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0628
fit σ = 0.0062
rank = 13
SiPM 02 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0631
fit σ = 0.0065
rank = 3
SiPM 03 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0630
fit σ = 0.0070
rank = 6
SiPM 04 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0629
fit σ = 0.0065
rank = 11
SiPM 05 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0630
fit σ = 0.0063
rank = 7
SiPM 06 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0629
fit σ = 0.0066
rank = 12
SiPM 07 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0628
fit σ = 0.0061
rank = 15
SiPM 08 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0651
fit σ = 0.0106
rank = 1
SiPM 09 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0629
fit σ = 0.0072
rank = 10
SiPM 10 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0630
fit σ = 0.0064
rank = 8
SiPM 11 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0630
fit σ = 0.0062
rank = 9
SiPM 12 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0631
fit σ = 0.0068
rank = 4
SiPM 13 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0631
fit σ = 0.0063
rank = 5
SiPM 14 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0628
fit σ = 0.0063
rank = 14
SiPM 15 fraction of full 16-SiPM sum
Run 171, 1.259 V
fit mean = 0.0631
fit σ = 0.0070
rank = 2
Events
fraction

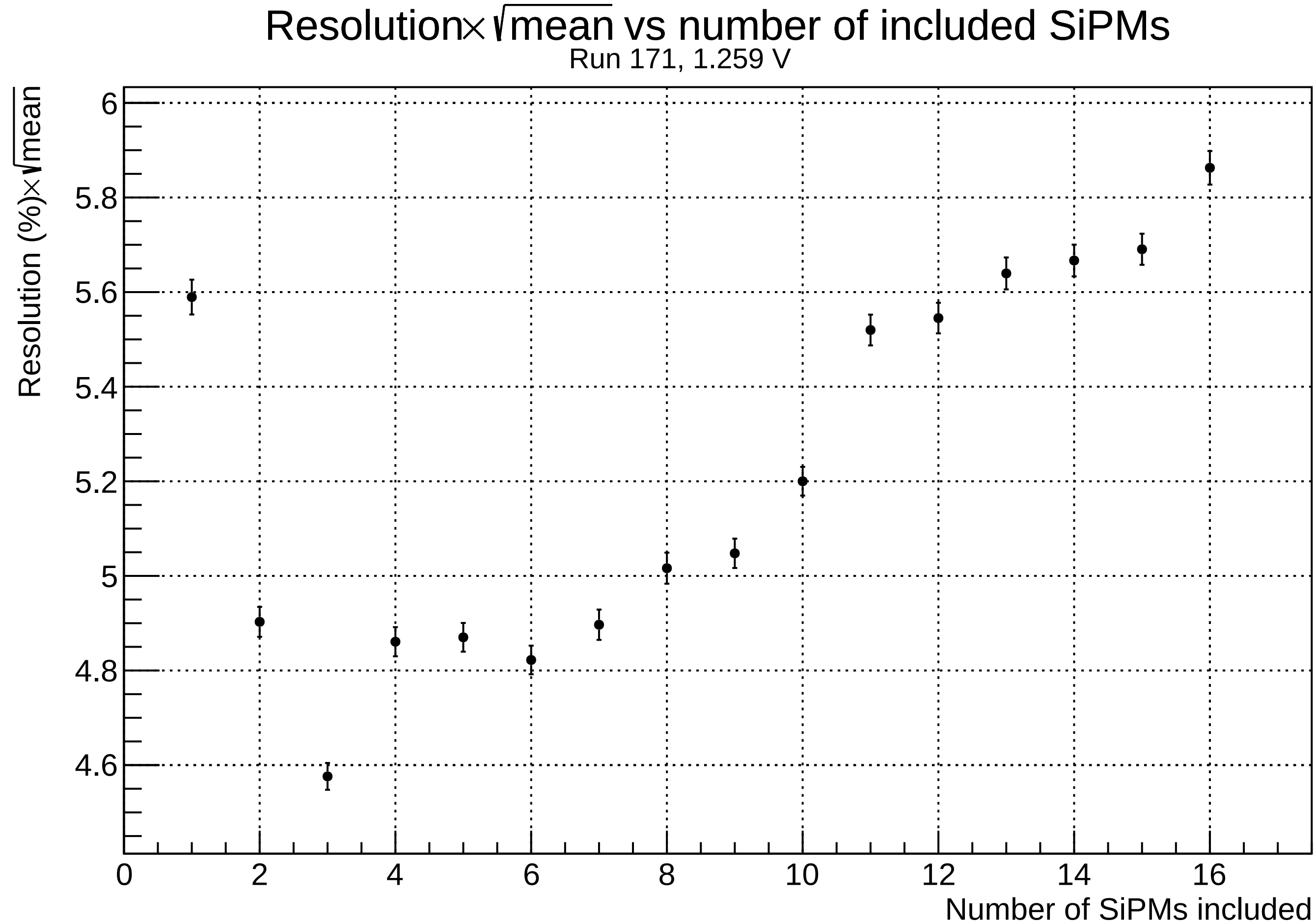

Fit mean fraction collected by each SiPM
Run 171, 1.259 V
Ideal equal share (1/16)
Fit mean fraction of full 16-SiPM sum
0.066
0.064
0.062
0.06
0.058
0.056
0.054
0.052
0
2
4
6
8
10
12
14
16
SiPM

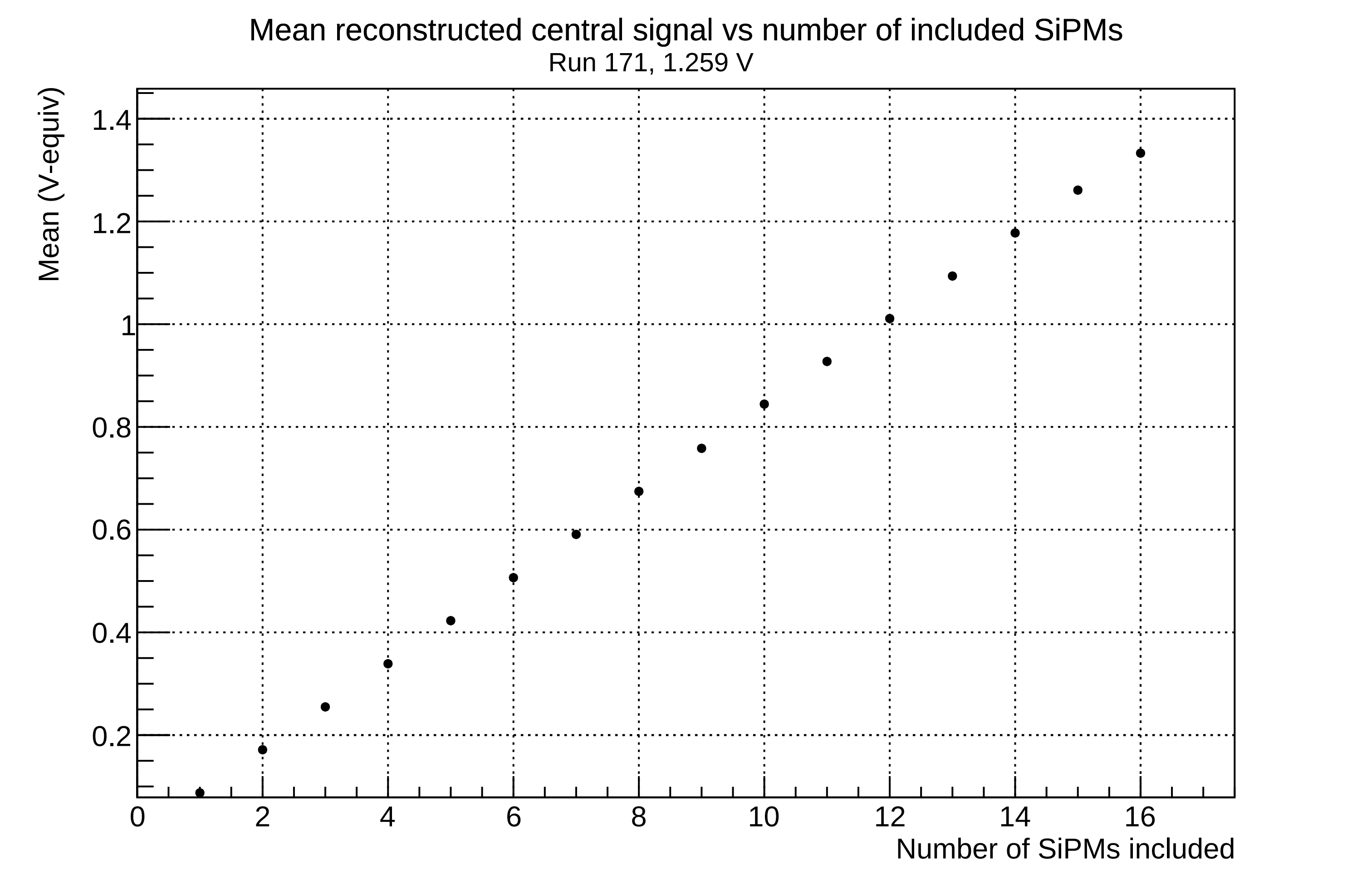

Fit mean fraction ranked
Run 171, 1.259 V
Fit mean fraction
0.066
0.064
0.062
0.06
0.058
0.056
0.054
0.052
0
2
4
6
8
10
12
14
16
Rank
Ideal equal share (1/16)

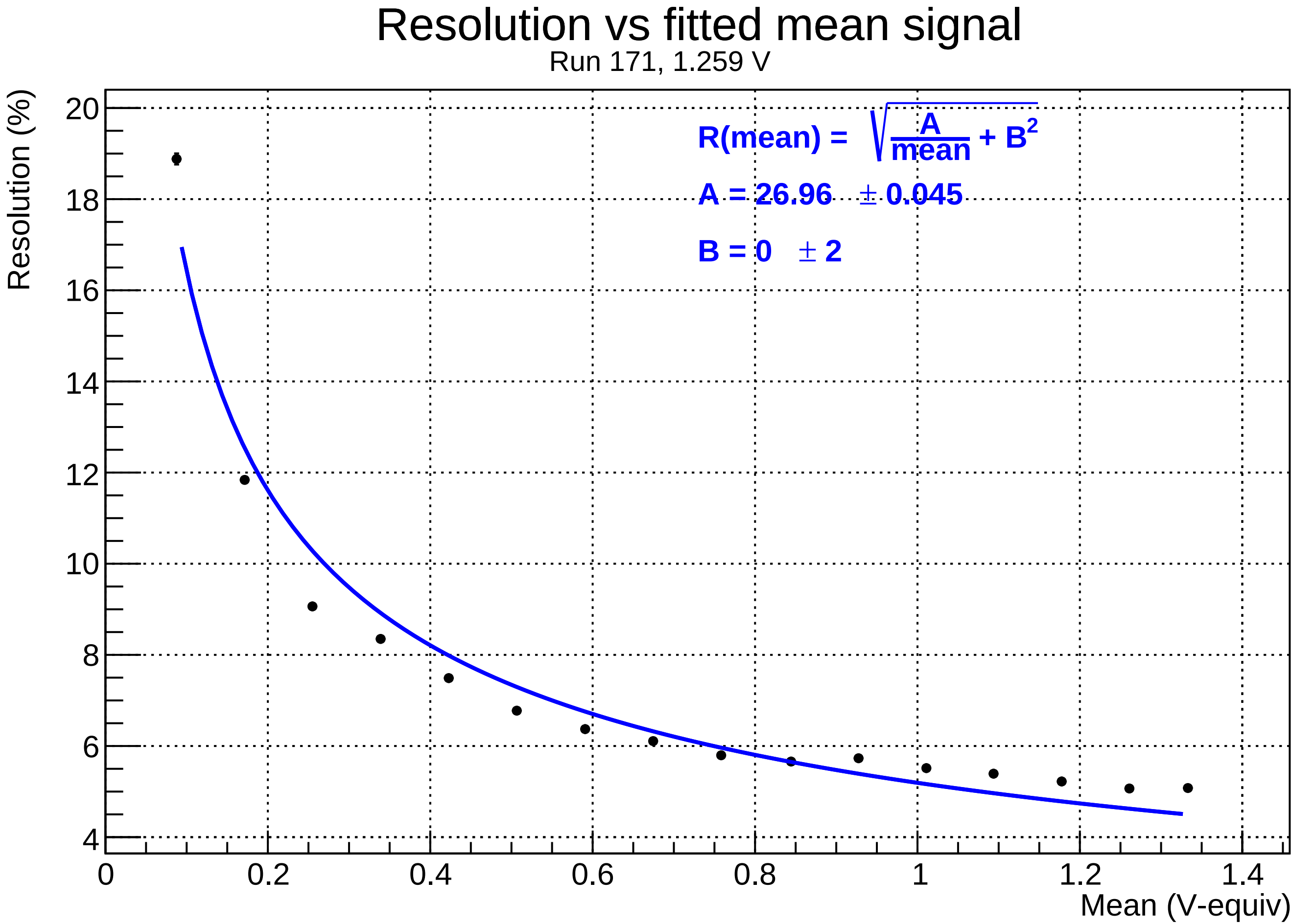

Fit mean cumulative fraction vs number of SiPMs
Run 171, 1.259 V
Fit mean collected fraction
Number of SiPMs included
0
0.2
0.4
0.6
0.8
1
0
2
4
6
8
10
12
14
16



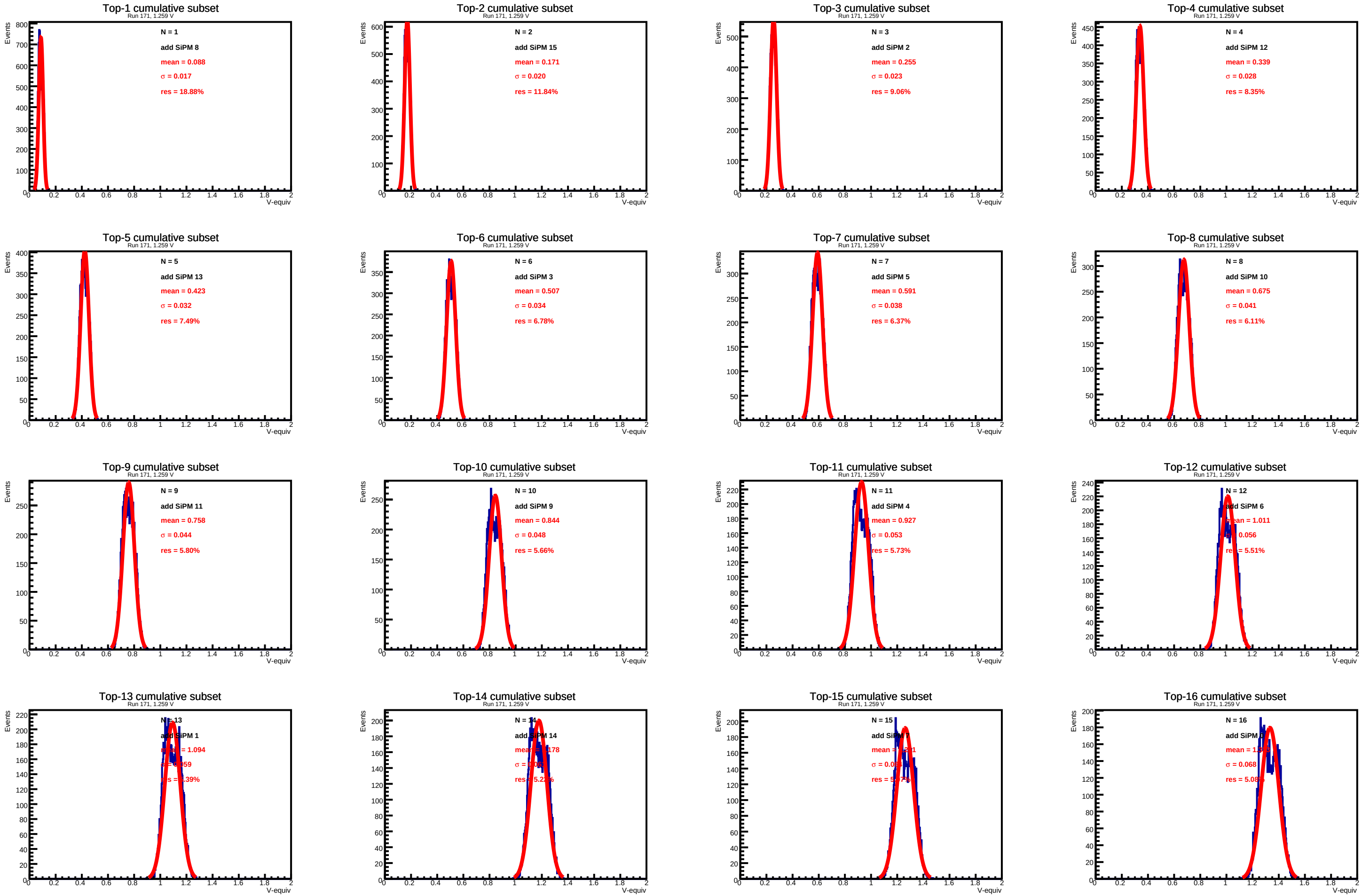
Resolution vs number of included SiPMs
Run 171, 1.259 V
Resolution (%)
Number of SiPMs included
$R(mean) = \sqrt{\frac{A}{mean} + B^2}$
A = 26.96 ± 0.045
B = 0 ± 2
R^2 = 0.9685
χ^2/NDF = 169.093


Resolution × √mean vs number of included SiPMs
Run 171, 1.259 V
Resolution (%) × √mean
Number of SiPMs included
6
5.8
5.6
5.4
5.2
5
4.8
4.6
0
2
4
6
8
10
12
14
16

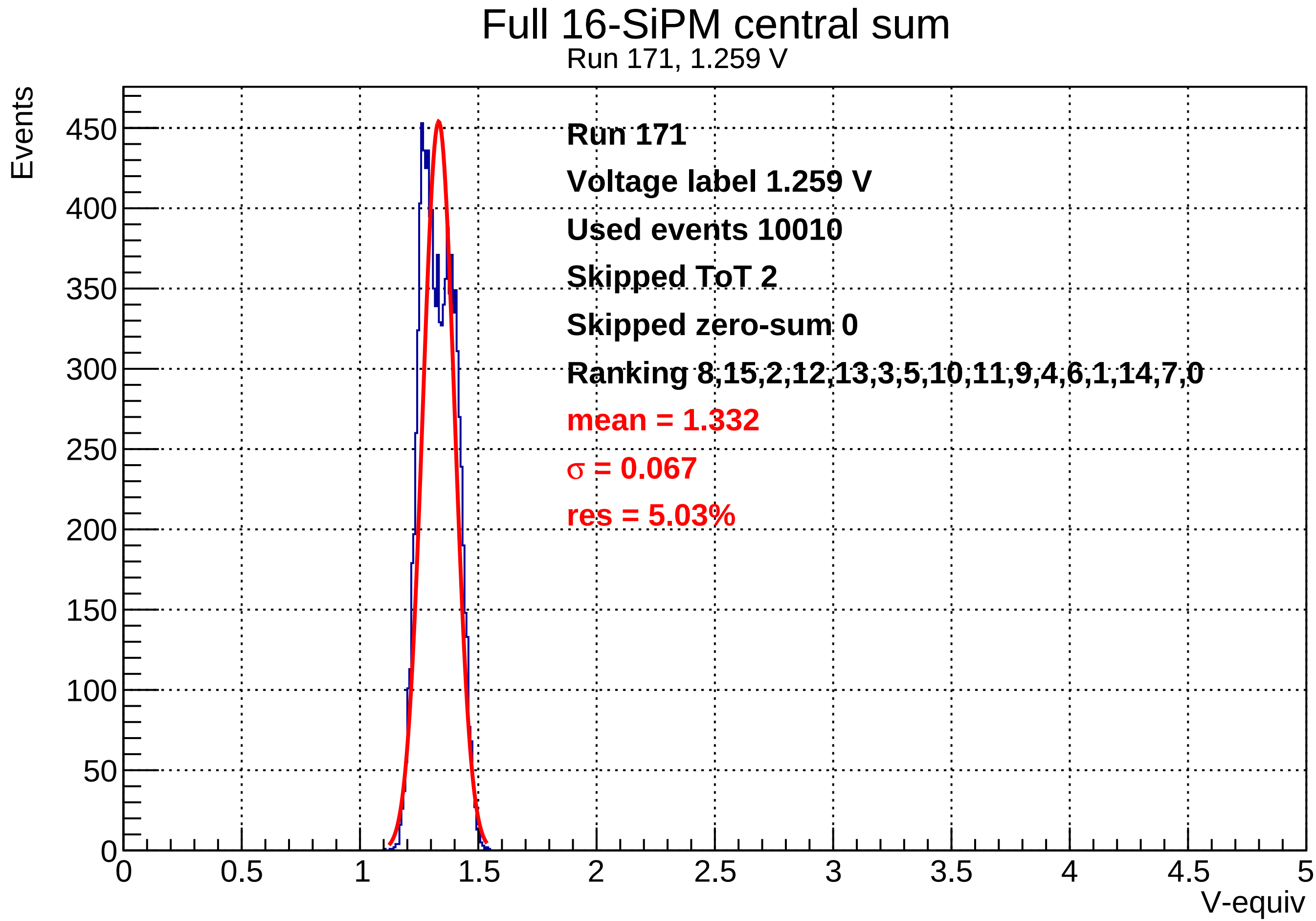

Mean reconstructed central signal vs number of included SiPMs
Run 171, 1.259 V
Mean (V-equiv)
1.4
1.2
1
0.8
0.6
0.4
0.2
0
2
4
6
8
10
12
14
16
Number of SiPMs included



Resolution vs fitted mean signal
Run 171, 1.259 V
R(mean) = $\sqrt{\frac{A}{mean} + B^2}$
A = 26.96 ± 0.045
B = 0 ± 2
Resolution (%)
Mean (V-equiv)
20
18
16
14
12
10
8
6
4
0
0.2
0.4
0.6
0.8
1
1.2
1.4



Top-1 cumulative subset
Run 171, 1.259 V
N = 1
add SiPM 8
mean = 0.088
σ = 0.017
res = 18.88%
Top-2 cumulative subset
Run 171, 1.259 V
N = 2
add SiPM 15
mean = 0.171
σ = 0.020
res = 11.84%
Top-3 cumulative subset
Run 171, 1.259 V
N = 3
add SiPM 2
mean = 0.255
σ = 0.023
res = 9.06%
Top-4 cumulative subset
Run 171, 1.259 V
N = 4
add SiPM 12
mean = 0.339
σ = 0.028
res = 8.35%
Top-5 cumulative subset
Run 171, 1.259 V
N = 5
add SiPM 13
mean = 0.423
σ = 0.032
res = 7.49%
Top-6 cumulative subset
Run 171, 1.259 V
N = 6
add SiPM 3
mean = 0.507
σ = 0.034
res = 6.78%
Top-7 cumulative subset
Run 171, 1.259 V
N = 7
add SiPM 5
mean = 0.591
σ = 0.038
res = 6.37%
Top-8 cumulative subset
Run 171, 1.259 V
N = 8
add SiPM 10
mean = 0.675
σ = 0.041
res = 6.11%
Top-9 cumulative subset
Run 171, 1.259 V
N = 9
add SiPM 11
mean = 0.758
σ = 0.044
res = 5.80%
Top-10 cumulative subset
Run 171, 1.259 V
N = 10
add SiPM 9
mean = 0.844
σ = 0.048
res = 5.66%
Top-11 cumulative subset
Run 171, 1.259 V
N = 11
add SiPM 4
mean = 0.927
σ = 0.053
res = 5.73%
Top-12 cumulative subset
Run 171, 1.259 V
N = 12
add SiPM 6
mean = 1.011
0.056
5.51%
Top-13 cumulative subset
Run 171, 1.259 V
N = 13
add SiPM 1
1.094
Top-14 cumulative subset
Run 171, 1.259 V
Top-15 cumulative subset
Run 171, 1.259 V
N = 15
Top-16 cumulative subset
Run 171, 1.259 V
N = 16
σ = 0.068
Events
V-equiv


Full 16-SiPM central sum
Run 171, 1.259 V
Events
V-equiv
0
50
100
150
200
250
300
350
400
450
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5
Run 171
Voltage label 1.259 V
Used events 10010
Skipped ToT 2
Skipped zero-sum 0
Ranking 8,15,2,12,13,3,5,10,11,9,4,6,1,14,7,0
mean = 1.332
σ = 0.067
res = 5.03%